

Levee with Toe Drain

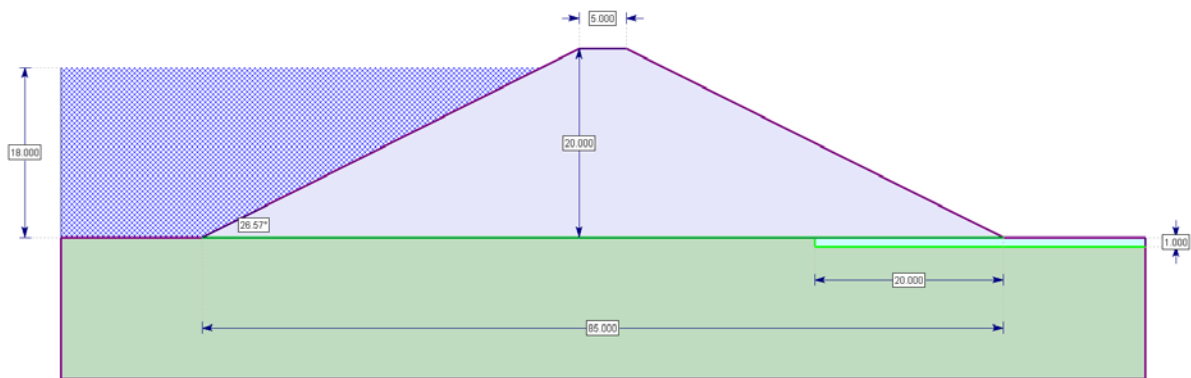
In this tutorial, finite element groundwater seepage analysis is used to simulate a levee with a horizontal toe drain. Toe drains are often used to prevent capillary rise on the downstream sloping surface. *Phase²* can be used to test the effectiveness of different drain configurations.

The finished tutorial can be found in the **Tutorial 21 Levee with Toe Drain.fez** file. All tutorial files installed with *Phase²* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase²* main menu.

Topics Covered

- Seepage analysis
- Levee drainage
- Multiple materials
- Phreatic surface
- Groundwater only calculation
- Flownets

Geometry



Model

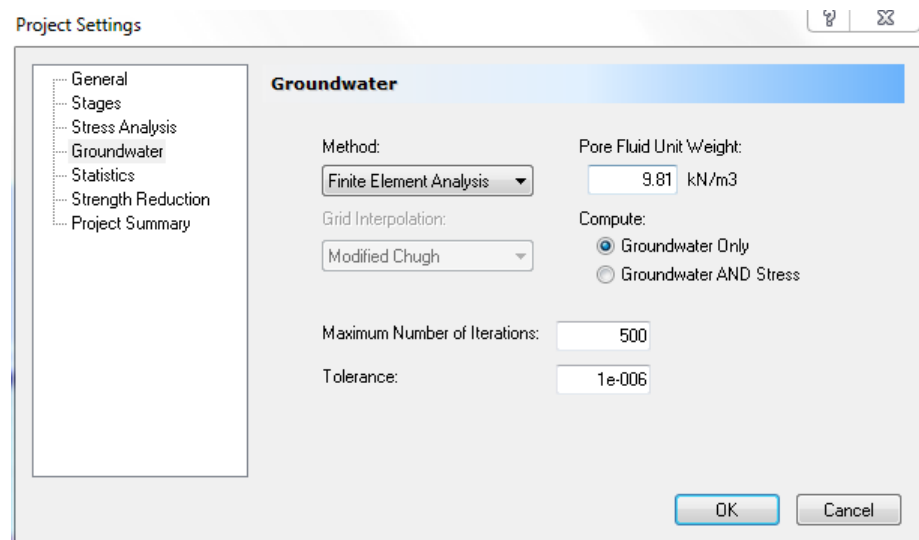
Start the *Phase²* Model program.

Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Set the Units of Measurement to “Metric, stress as kPa”.

Click the Groundwater tab. Under Method choose Finite Element Analysis. In this tutorial we are not interested in performing a stress analysis, so under Compute, select Groundwater Only. Close the Project Settings dialog by pressing the OK button.



Boundaries



First add the external boundary. Select the **Add External** option in the **Boundaries** menu and enter the following coordinates:

0 , 0

36 , 18 *

40 , 20

* This point is required to specify the height of the ponded water later in the tutorial.

45 , 20

85 , 0

100 , 0

100 , -15

-15 , -15

-15 , 0

c (to close the boundary)



Hit Enter to finish entering points. This defines the external boundary, which includes the levee sitting on top of low permeability soil (see figure at the start of the tutorial). Select **View** → **Zoom** → **Zoom All** to center and maximize the model in the view.



Now we need to add material boundaries. Firstly, we will define the boundary between the levee and the underlying soil. Go to the **Boundaries** menu and select **Add Material**. Enter the following points:

0 , 0

85 , 0

Hit Enter to finish entering points.

TIP: when you are entering boundary points, the cursor should snap to existing points. Therefore you do not need to type in coordinates if a point already exists at that location. If your cursor does not snap to existing points, right click with the mouse when you are creating a boundary and select Snap in the popup menu to turn on the snapping option.

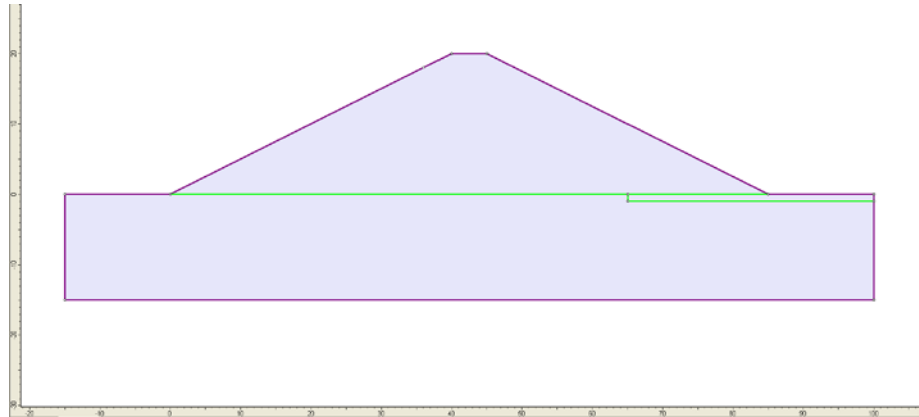
To define the toe drain material boundary, add another material boundary with the following coordinates:

100 , -1

65 , -1

65 , 0

Hit Enter. Your model should now look like this:



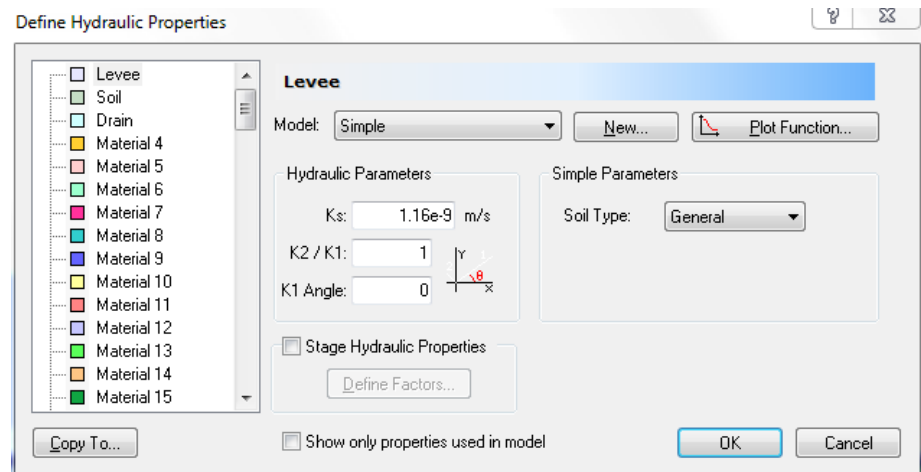
Material Properties



Select **Define Materials** from the **Properties** menu. You will see the default material properties for Material 1. In this tutorial we don't care about the strength or stiffness of the solid material therefore leave all the default values. Change the name of Material 1 to Levee. Now click on the Material 2 tab. Change the name of Material 2 to Soil. Similarly, change the name of Material 3 to Drain. Click OK to close the dialog.



We now need to define the fluid flow properties of the soil. Go to the **Properties** menu and choose **Define Hydraulic**. Click on the Levee tab at the top of the dialog. Enter $1.16\text{e-}9$ for K_s . Leave all other values as the default values as shown.



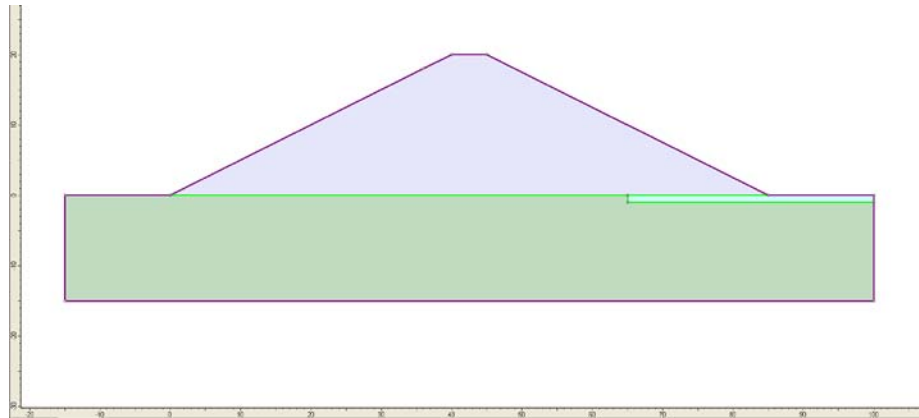
The underlying soil is assumed to be essentially impermeable, so click on the Soil tab and enter a value for K_s of $1.0\text{e-}20$ m/s.

For the Drain material, enter a permeability of $1\text{e-}6$ m/s to simulate a high permeability sand drain.

Assign Material Properties



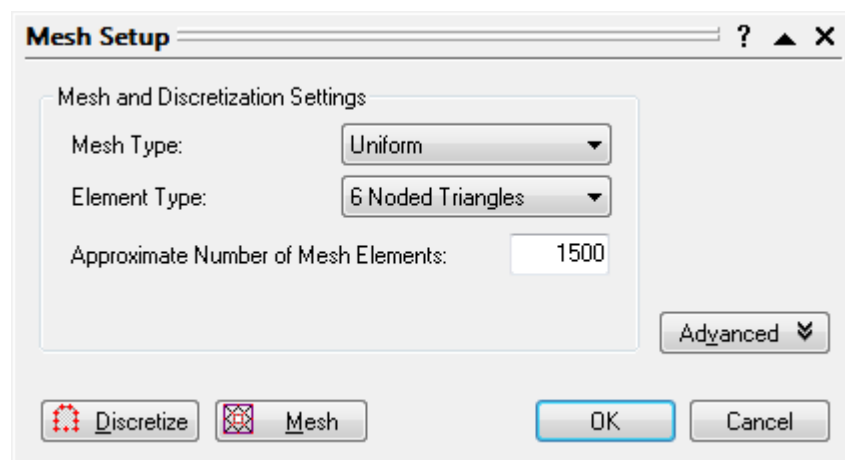
By default, the entire model is assigned the properties of Levee (material 1). To assign the correct material properties to the different parts of the model, go to the **Properties** menu and select **Assign Properties**. Select Soil from the Assign dialog and click near the bottom of the model. Now select Drain and click inside the drain region (the narrow rectangle at the toe of the levee). Alternatively you can assign material by simply right-clicking inside the region of interest and choosing Assign Material. Your model should now look like this:



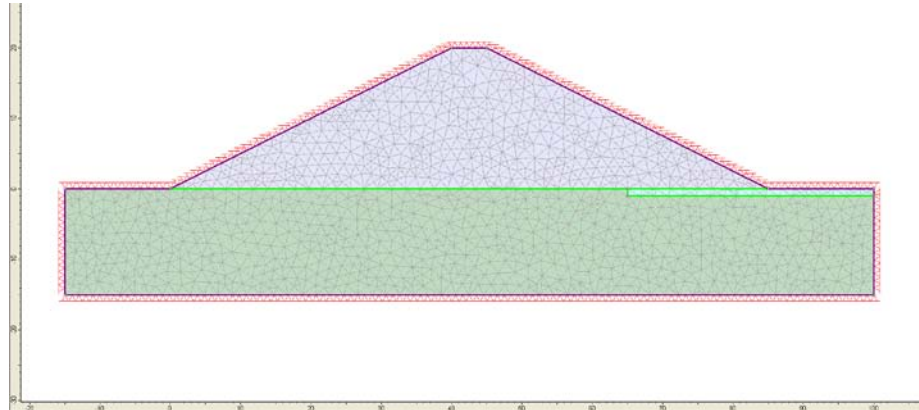
Mesh



Now generate the finite element mesh. Select the **Mesh Setup** option in the **Mesh** menu. Set the Mesh Type to Uniform. Leave the default number of elements (1500) but set the Element Type to 6 Noded Triangles. Here we wish to use 6-noded triangles to get more degrees of freedom in the narrow drain region. Click the Discretize button followed by the Mesh button.



Close the Mesh Setup dialog by selecting the OK button. Your model should now appear as shown.



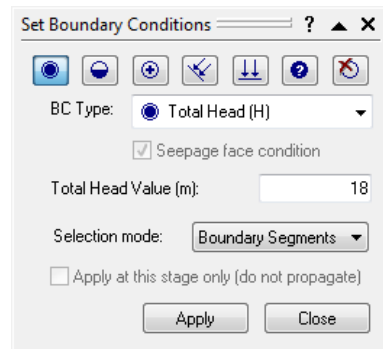
Boundary Conditions



To set the boundary conditions, we first need to display the groundwater boundary conditions. From the **Groundwater** menu, select **Show Boundary Conditions**.



The model shows the default boundary conditions (no flow across any external boundary segment). We wish to simulate ponded water to the left of the levee and a seepage face on the right. The ponded water has a depth of 18 m, therefore we will set the total head for these boundaries to 18 m. To do this, choose **Set Boundary Conditions** from the **Groundwater** menu. For BC Type choose Total Head. Enter a Total Head Value of 18.



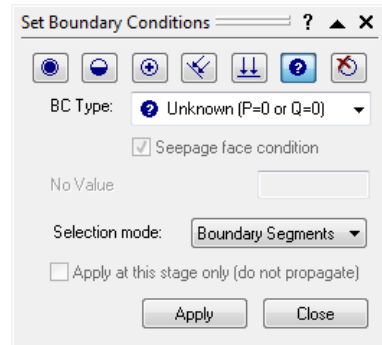
Now select the two boundary segments that enclose the ponded water:

Line 1: from (-15,0) to (0,0)

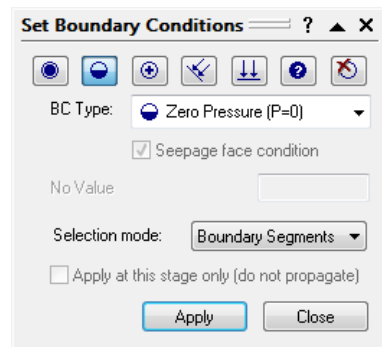
Line 2: from (0,0) to (36,18)

Click Apply.

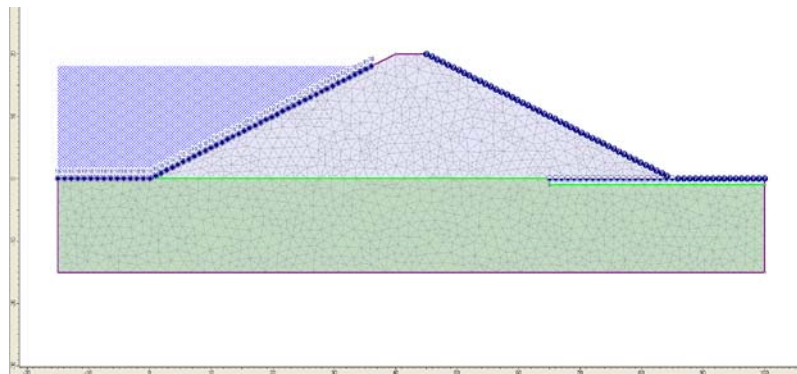
To set up the seepage face on the right of the levee, choose Unknown for the BC Type. Click on the right slope of the levee (line from 45,20 to 85,0) and the horizontal surface at the right (line from 85,0 to 100,0). Click Apply.



We will assume that the drain provides a drained boundary such that the pressure along the top of the drain is 0. Therefore choose Zero Pressure for the BC Type.



Click on the top of the drain material (line from 65,0 to 85,0) and click apply. Close the dialog and your model should look like this:



TIP: you can also right-click on a boundary to define its boundary conditions.



You have now completed the definition of the model. Save the model using the **Save** option in the **File** menu.

Compute



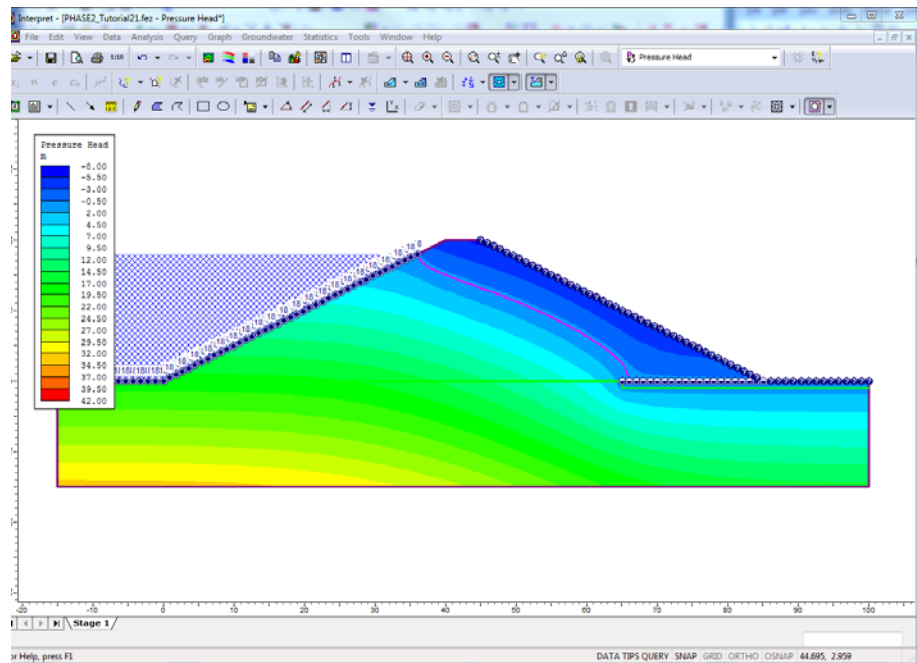
Since we are only interested in the groundwater results, we only need to run the groundwater computation. Select **Compute (groundwater only)** from the **Groundwater** menu (or click the Compute groundwater button in the toolbar). The analysis should take a few seconds to run.



Once the model has finished computing (Compute dialog closes), select **Interpret** in the **Analysis** menu to view the results.

Interpret

After you select the Interpret option, the Interpret program starts and reads the results of the analysis. A screen is displayed showing the pressure head results. Your plot should look like this:



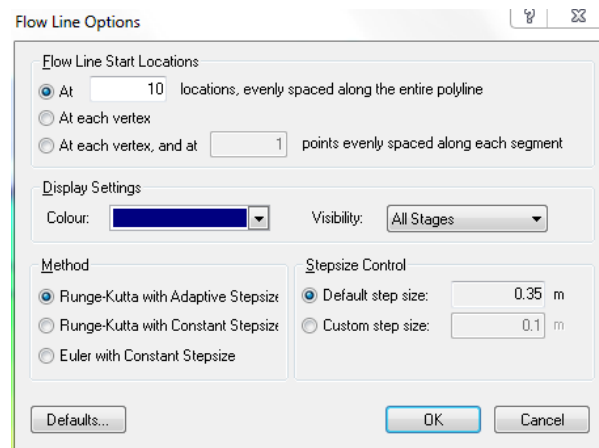
The purpose of the toe drain was to prevent the phreatic surface from intersecting the right (downstream) side of the levee. The phreatic surface is shown as a pink line on the plot and it is clear that it does not intersect the boundary, meaning that the drain performed as desired.



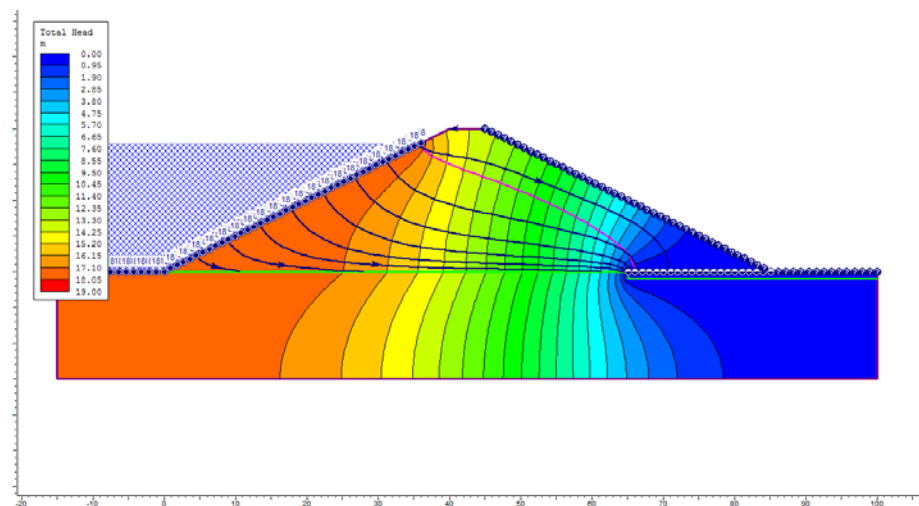
We can easily construct a flow net to examine the results in more detail. Change the quantity being plotted from Pressure Head to Total Head using the drop down menu on the tool bar. Now right-click on the model and select Contour Options. Under Mode select **Filled (with lines)** and then select Done. You will now see the equipotential lines of the flownet.



To plot the flow lines, go the **Groundwater** menu and select **Add Multiple Flow Lines**. Select the top left corner of the levee as the first point (40,20). Now select the bottom left corner of the levee (0,0). Hit enter to finish. You will see the Flow Line Options dialog. Here you can choose how many flow lines you wish to plot. Under Flow Line Start Locations select the first option and leave the default value (10 locations, evenly spaced along the polyline).



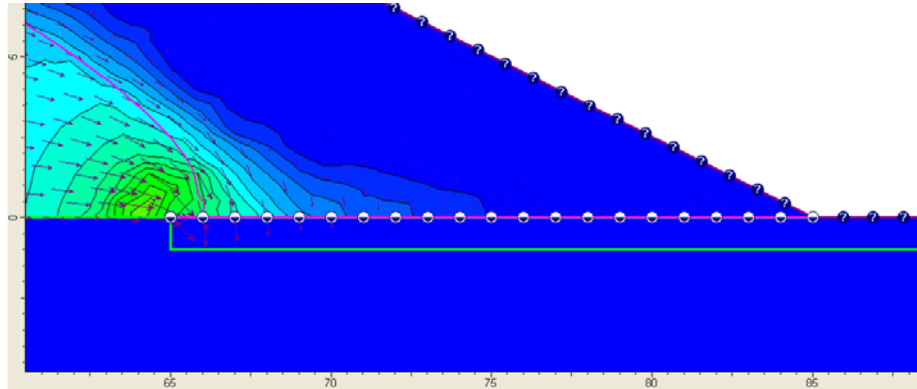
Click OK to close the dialog. You will now see 10 flow lines plotted as shown.



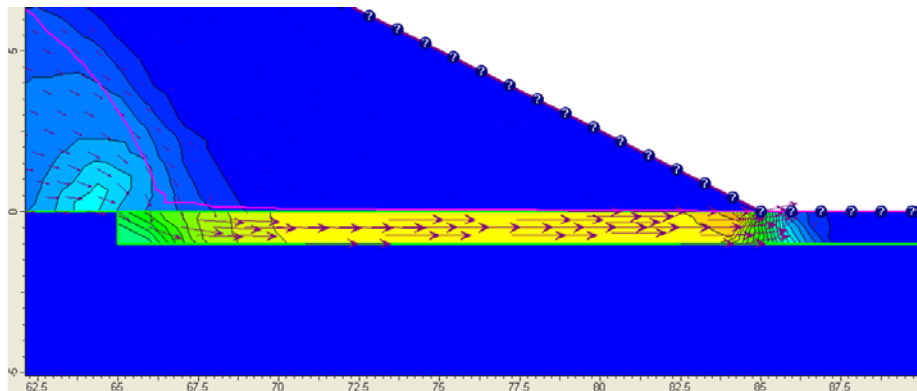
This concludes the Levee with Toe Drain tutorial.

Modeling Comments

If you display the flow vectors for this model, and view the discharge velocity contours (see figure below), you will observe that there is apparently no flow taking place within the drain material. This is because the zero pressure boundary condition along the top of the drain, acts as a “sink”, and this is what simulates the drainage condition. The high permeability of the drain material does not create the drainage condition, in this case.

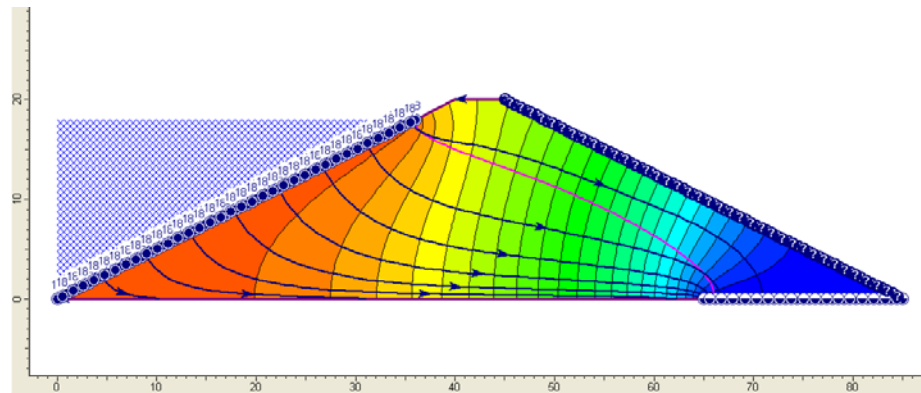


However, if you remove the zero pressure boundary condition at the top of the drain, and re-run the analysis, you will then see actual flow through the drain material, as shown in the figure below. This is due to the difference in permeability of the drain and levee materials.



For this particular model, the analysis results (pressure head, total head, location of water table) are very similar, with or without the boundary condition. However, this will not always be the case, and in general it is recommended that the zero pressure boundary condition is used to enforce the drainage condition at the desired location.

Another modeling alternative is to exclude the base and drain material altogether, and just model the levee material with boundary conditions, as shown in the next figure.

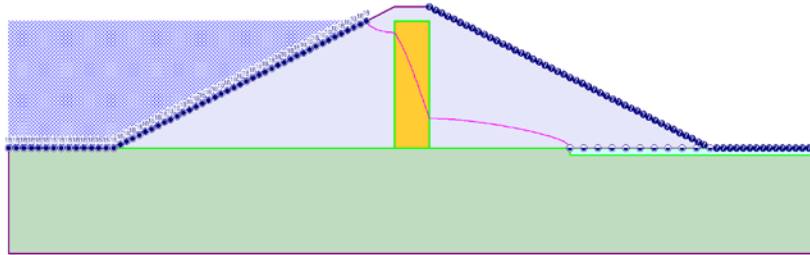


If you are only interested in groundwater results, and the base material is assumed to be impermeable, then it is sufficient to only model the levee as shown in the above figure.

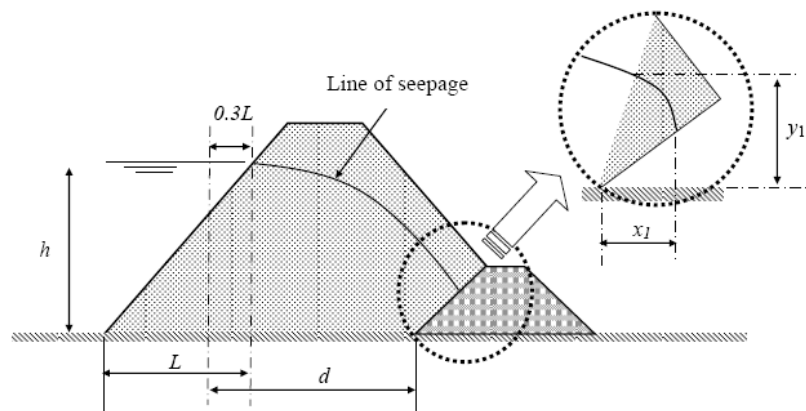
NOTE: one reason you may wish to include the base material in the model, is that a slope stability analysis can be easily carried out on the entire model, if desired.

Additional Exercises

We can simulate a levee with a low permeability core by specifying material boundaries to define the core and setting up a new material with a lower permeability (say $1\text{e-}11$ m/s). An example is shown below:



Another possibility is to construct a levee with a non-horizontal toe drain as shown.



This type of model is described in Groundwater Verification Problem #4.